

Komplexe Zahlen

David Seez

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- 1 Motivation
- 2 Definition und algebraische Struktur
- 3 Geometrie
- 4 Polardarstellung
- 5 Ausblick

Motivation und Geschichte

Quadratische Gleichungen

If some one say: "You divide ten into two parts: multiply the one by itself; it will be equal to the other

(48)

taken eighty-one times."* Computation: You say, ten less thing, multiplied by itself, is a hundred plus a (35) square less twenty things, and this is equal to eighty-one things. Separate the twenty things from a hundred and a square, and add them to eighty-one. It will then be a hundred plus a square, which is equal to a hundred and one roots. Halve the roots; the moiety is fifty and a half. Multiply this by itself, it is two thousand five hundred and fifty and a quarter. Subtract from this one hundred; the remainder is two thousand four hundred and fifty and a quarter. Extract the root from this; it is forty-nine and a half. Subtract this from the moiety of the roots, which is fifty and a half. There remains one, and this is one of the two parts.

$$10 = x + y$$

Wörtlich zitiert und übersetzt in [al-Khwārizmī, *The Algebra of Mohammed ben Musa*, 1831], S. 48.

Original: [al-Khwārizmī, *Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wa-l-Muqābala*, 820]

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$$y^2 = 81x$$

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Für uns: $y = 10 - x$

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Gelöst mit heutiger p-q formel

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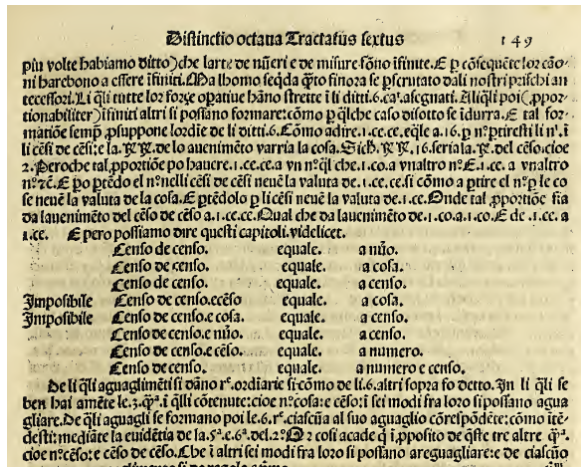
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Cubische/Quartische Gleichungen



[Pacioli, *Summa de Arithmetica, Geometria, Proportioni et Proportionalità*,

1494]



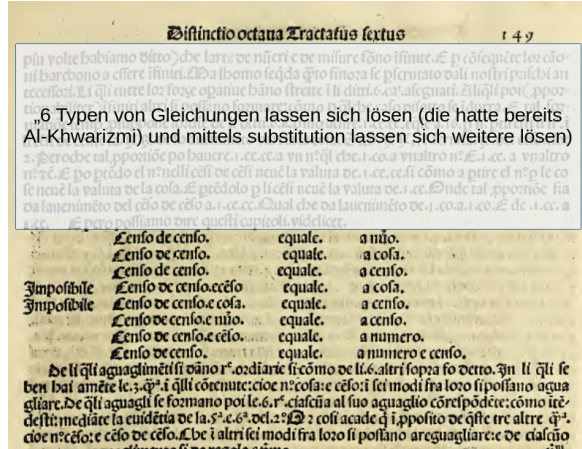
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Distinctio octava Tractatus sextus

149

„6 Typen von Gleichungen lassen sich lösen (die hatte bereits Al-Khwarizmi) und mittels substitution lassen sich weitere lösen)

möglich	Quadrat des Quadrats	gleich	Zahl
möglich	Quadrat des Quadrats	gleich	Unbekannte
möglich	Quadrat des Quadrats	gleich	Quadrat
unmöglich	Quadrat des Quadrats plus Quadrat	gleich	Unbekannte
unmöglich	Quadrat des Quadrats plus Unbekannte	gleich	Quadrat
möglich	...	gleich	...
möglich	...	gleich	...
möglich	...	gleich	...

De li qli aguagliamēti li dano r^o ordiane si cōmo de li.6.altri sopra fo detto. In li qli le ben hai amēte le.3. q^o. i qli cōtēnute: cioe n^o: cofa: e cēfo: i sei modi fra loro si possano aguagliare. De qli aguagli le formano poi le.6.r^o. ciaschūa al suo aguaglio cōrēspōdēre: cōmo itē. dēsti: mediāte la euidētia de la.5^a. e.6^a. del.2^o. De 2 colli acade q^o i pposito de qste tre altre q^o. cioe n^o: cēfo: e cēfo de cēfo. E be i altri sei modi fra loro si possano areguagliare: e de ciaschūo

Motivation und Geschichte

Cubische/Quartische Gleichungen

DE ARITHMETICA. LIB. X.

31

De cubo aequali rebus & numero.

Cap. XII.

DEMONSTRATIO.

Sit etiam cubus aequalis rebus & numero, & sint duo cubi $d c \& d f$, quorum latera $a \& b c$, producit tertiam partem numeri rerum, inuicem ducta, & ipsi cubi iuncti aequales illi numero, dico $a c$ esse rei quę sitæ æstimationē, cum enim $ex a b$, in $b c$, fiat tertia pars numeri rerum, $ex a b$ in $b c$ ter, fiet numerus rerū, & $ex a c$ in productum $ex a b$ in $b c$ ter, fiet res ipse, posita $a c$ re, at $ex a c$ in productum $a b$ in $b c$ ter, sunt sex corpora, quorum tria sunt $ex a b$ in quadratum $b c$, alia tria $ex b c$ in quadratum $a b$, hæc igitur sex corpora, æqualia sunt rebus, ipsa uero cum cubis $d c \& d f$, ex primo supposito capituli sexti constituunt cubum $a b$, cubi etiam $d c \& d f$, æquivalent numero proposito, igitur cubus $a b$, æqualis est rebus & numero propositis, quod erat demonstrandum, si perest ostendere, quod triplum $a c$ in productum $a b$ in $b c$, sit æquale sex corporibus, id ostendā, si probauero $ex a b$, in $b c$ ducto in $a c$, fieri duo corpora $ex a b$ in quadratum $b c$, & $ex b c$ in quadratum $a b$, nam quod fit $ex a c$ in productum $a b$ in $b c$, æquale est ei, quod fit $ex a b$ in superficiem $b c$, latera enim omnia omnibus sunt æqualia, sed hoc æquale est ei, quod fit $ex a b$ in $c d \& d f$, quod autem fit $ex a b$ in $d f$, æquale est ei, quod fit $ex c b$ in quadratum $a b$, quoniam latera omnia omnibus sunt æqualia, quod igitur $ex a c$, in productum $a b$ in $b c$ ter, æquale est his, quę sunt $ex a b$ in quadratum $b c$ & $ex b c$ in quadratum $a b$, quod est propositum.

REGULA.

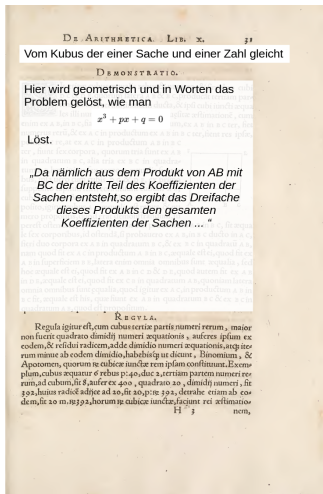
Regula igitur est, cum cubus tertie partis numeri rerum, maior non fuerit quadrato dimidij numeri æquationis, auferes ipsum ex eodem, & residui radicem, adde dimidio numeri æquationis, atque iterum minue ab eodem dimidio, habebis q̃ ut dicunt, Binomium, & Apotomen, quorum se cubice iunctæ rem ipsam constituunt. Exemplum, cubus æquatur 6 rebus p. 40, duc 2, tertiam partem numeri rerum, ad cubum, fit 8, aufer ex 400, quadrato 20, dimidij numeri, fit 392, huius radicē adijce ad 20, fit 26, p. 19 392, deuiat etiam ab eodem, fit 20 m. 19 392, horum se cubice iunctæ, faciunt rei æstimationem,

H 3 nem,



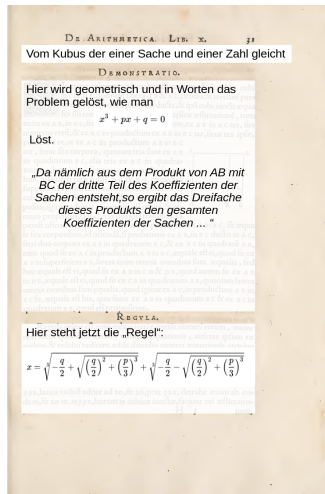
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Cubische/Quartische Gleichungen



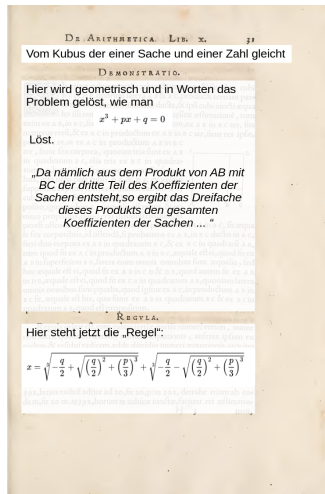
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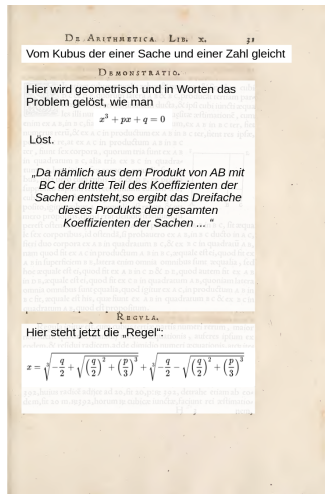


$$x^3 + px + q = 0$$

$$\Leftrightarrow x = \sqrt[3]{-\frac{q}{2} + \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}}$$

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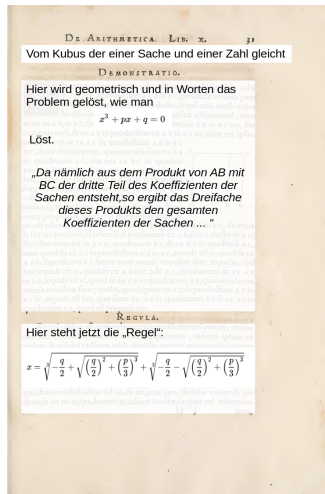
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$$x^3 - 15x - 4 = 0 \quad \Leftrightarrow \quad x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}}$$

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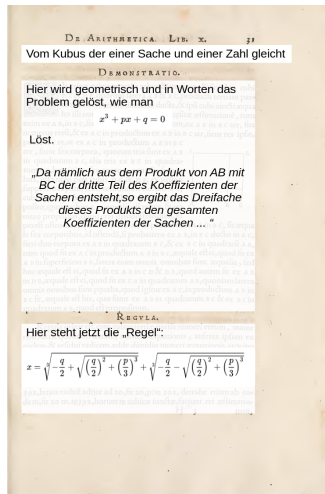
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François Viète [1540-1603] Führt Rechnen mit Symbolen ein (+, -, aequabitur) Außerdem eine Formel für eine Nullstelle bei Quartiken.

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René Descart [1596-1650-] rechnet mit negativen Wurzeln!

Definition

Hamiltonsche Konstruktion [Hamilton, „Theory of Conjugate Functions, or Algebraic Couples; with a Preliminary and Elementary Essay on Algebra as the Science of Pure Time“, 1837]

Definition 1 (Komplexe Zahlen als $\mathbb{R}^2$)

Setze $\mathbb{C} := \mathbb{R}^2$ mit komponentenweiser Addition und

$$\mathbf{1} := (1, 0), \quad \mathbf{1}^2 = \mathbf{1}, \quad \mathbf{i} := (0, 1), \quad \mathbf{i}^2 = -\mathbf{1}.$$

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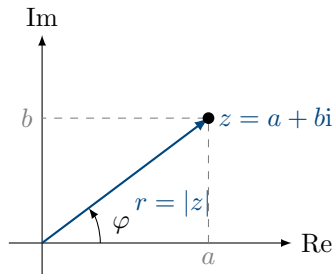
Die definition der Multiplikation folgt aus Distributivität:

$$\begin{aligned}(a\mathbf{1} + b\mathbf{i})(c\mathbf{1} + d\mathbf{i}) &= ac\mathbf{1}^2 + (ad + bc)\mathbf{1}\mathbf{i} + bd\mathbf{i}^2 \\ &= ac\mathbf{1} + (ad + bc)\mathbf{i} - bd\mathbf{1} \\ &= (ac - bd)\mathbf{1} + (ad + bc)\mathbf{i}.\end{aligned}$$

$$\text{d. h. } (a, b) \cdot (c, d) := (ac - bd, ad + bc).$$

Die Gauß'sche Zahlenebene

Jede komplexe Zahl $z = a + bi$ entspricht einem Punkt $(a, b) \in \mathbb{R}^2$:

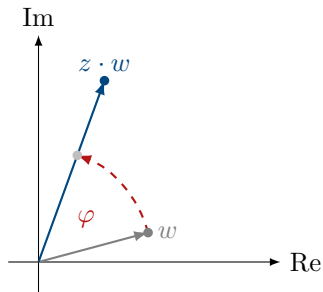


- Realteil: $\operatorname{Re}(z) = a$
- Imaginärteil: $\operatorname{Im}(z) = b$
- Betrag: $|z| = \sqrt{a^2 + b^2}$
- Argument: $\varphi = \operatorname{Arg}(z)$

Geometrie der Multiplikation

Drehstreckung

Multiplikation mit z wirkt auf w als **Drehstreckung**:



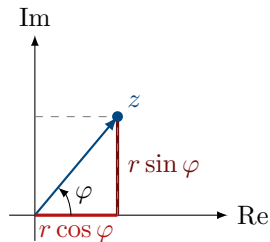
- **Drehung** um $\varphi = \text{Arg}(z)$
- **Streckung** um $r = |z|$

Formal: $|z \cdot w| = |z| \cdot |w|$,
 $\text{Arg}(z \cdot w) = \text{Arg}(z) + \text{Arg}(w)$.

Warum wirkt Multiplikation als Drehstreckung?

Schreibe $z = r(\cos \varphi + i \sin \varphi)$, $w = a + bi$:

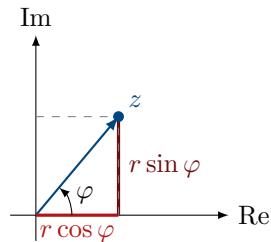
$$\begin{aligned} z \cdot w &= r(\cos \varphi + i \sin \varphi)(a + bi) \\ &= r[(a \cos \varphi - b \sin \varphi) \\ &\quad + i(a \sin \varphi + b \cos \varphi)]. \end{aligned}$$



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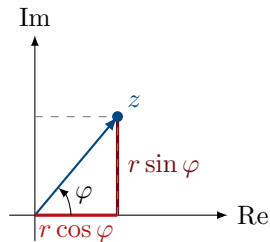
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Als Matrix auf $\mathbb{R}^2$:

$$z \cdot w \longleftrightarrow r \underbrace{\begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix}}_{R(\varphi)} \begin{pmatrix} a \\ b \end{pmatrix}.$$



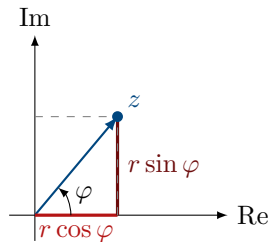
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$R(\varphi)$ = Drehmatrix um φ , r = Streckung.

Drehstreckungen $D_z: w \mapsto z \cdot w$ sind **$\mathbb{R}$ -lineare Abbildungen**. Daraus folgen die Körperaxiome direkt:

Assoziativität Siehe nächste Folie. ✓

Kommutativität Drehungen und Streckungen um den Ursprung vertauschen: $D_z \circ D_w = D_w \circ D_z$. ✓

Neutrales Element Die Identität $\text{id} = D_1$ (keine Drehung, keine Streckung). ✓

Inverses Jede Drehstreckung mit $r \neq 0$ ist invertierbar: $D_z^{-1} = D_{1/z}$. ✓

Distributivität D_z ist linear $\Rightarrow z(w_1 + w_2) = zw_1 + zw_2$. ✓

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$(\mathbb{C}, +, \cdot)$ ist ein **Körper** — und zwar ein *kommutativer*, weil Drehstreckungen um denselben Punkt stets vertauschen.

Zu $z = a + bi$ definiere die **Drehstreckungsmatrix**

$$\mathbb{C} \rightarrow \text{Mat}(2 \times 2, \mathbb{R}) \quad z \mapsto M_z := \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

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Schritt 1 — direkte Rechnung: $M_z \cdot M_w = M_{zw}$

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} c & -d \\ d & c \end{pmatrix} = \begin{pmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{pmatrix} = M_{zw}. \quad \checkmark$$

Assoziativität der Multiplikation

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Schritt 2 — Assoziativität vererbt sich: Da $z \mapsto M_z$ injektiv ist und $M_{zw} = M_z M_w$ gilt:

$$M_{z_1(z_2 z_3)} = M_{z_1} \underbrace{(M_{z_2} M_{z_3})}_{\text{Matrizen: assoz.}} = (M_{z_1} M_{z_2}) M_{z_3} = M_{(z_1 z_2) z_3}.$$

Also $z_1(z_2 z_3) = (z_1 z_2) z_3. \quad \checkmark$

